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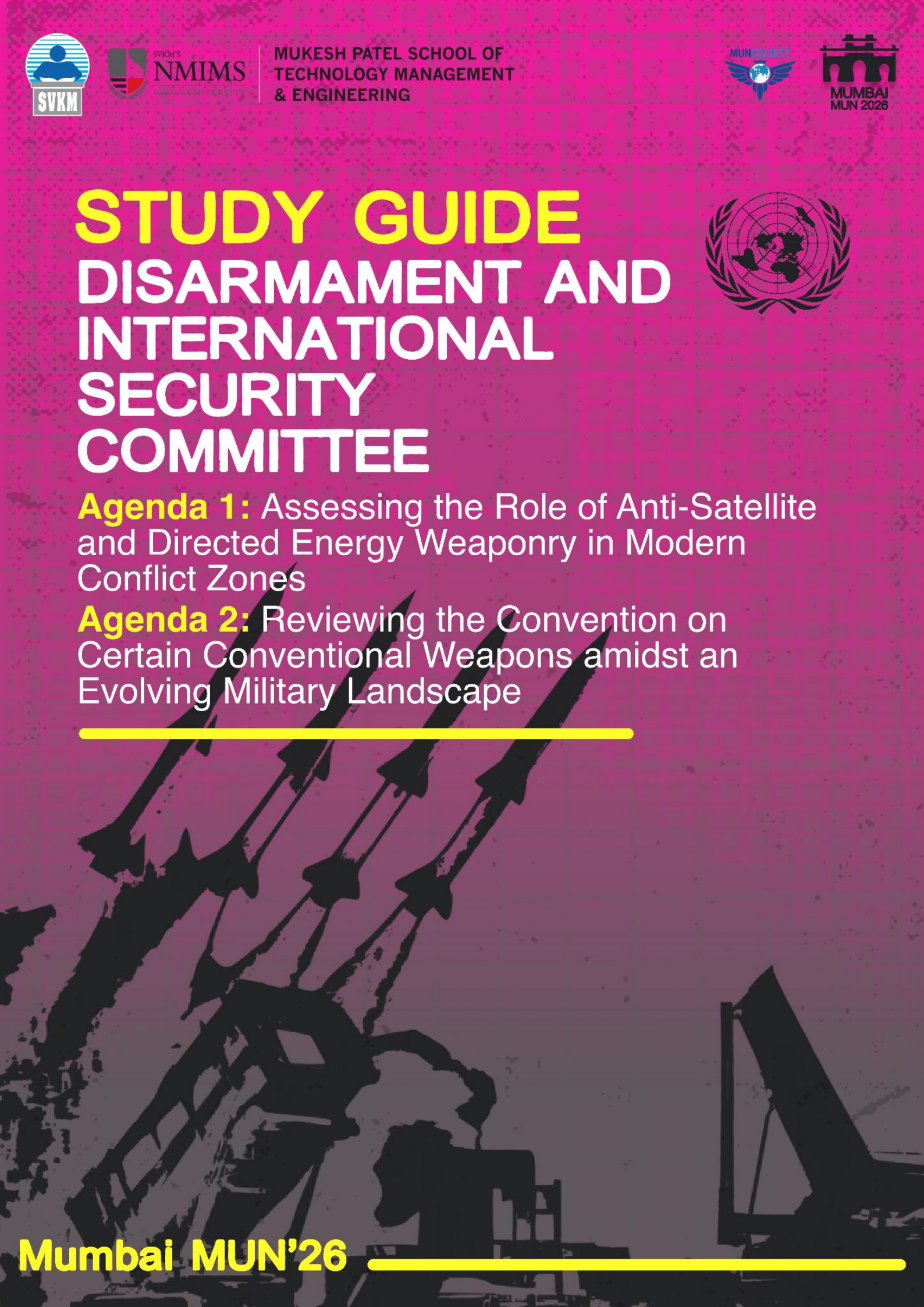
MUNSOCIETY
MUMBAI MUN 2026

STUDY GUIDE

DISARMAMENT AND INTERNATIONAL SECURITY COMMITTEE

Agenda 1: Assessing the Role of Anti-Satellite and Directed Energy Weaponry in Modern Conflict Zones

Agenda 2: Reviewing the Convention on Certain Conventional Weapons amidst an Evolving Military Landscape



Mumbai MUN'26

Letter from R&D

Dear Delegates,
The Department of Research & Development, MUN Society MPSTME, ecstatically welcomes all of you to the 13th edition of Mumbai MUN! Over the years, R&D has been guided by a singular ethos and vision: to make the Mumbai MUN conference a centre of meaningful discourse and debate on matters that rapidly shape the world we inhabit.

We are primarily tasked with formulating agendas and curating study guides that provide delegates with a comprehensive foundation before the conference. Our guides are the product of months of painstaking, human-driven research and production and are written with a *first-principles*, top-down approach.

Debate over matters of socio-political importance has taken centre stage in student life of late, with varying opinions and belief systems finding themselves continuously questioned. At Mumbai MUN 2026, as reflected by our theme "An Axiom Apart," we encourage delegates to bridge seemingly opposite truths and work together to creatively craft lasting solutions and partnerships.

With the *phare* that is our theme, our curations seek to foster meaningful and thought-provoking conversations on the topics that will define the global dynamics of the ever-changing world we live in. We seek to platform the perspectives that whittle away at the bureaucracy that barricades change.

The goal of any worthwhile MUN conference is to train the citizens of tomorrow to design, ideate, and create equitable world orders. The Research & Development team earnestly hopes Mumbai MUN 2026 will be a stepping stone for delegates to achieve that very goal.

Best Wishes,
Department of Research & Development
MUN Society MPSTME

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Agenda 1 - Assessing the Role of Anti-Satellite and Directed Energy Weaponry in Modern Conflict Zones

COMMITTEE OVERVIEW

The Disarmament and International Security Committee (DISEC) is the First Committee of the United Nations General Assembly. Established with the creation of the United Nations in 1945, it serves as the primary forum within the General Assembly for the consideration of disarmament and related international security questions. In accordance with the Charter of the United Nations, the General Assembly is mandated to consider the general principles of cooperation in the maintenance of international peace and security governing disarmament and regulations of armaments and to make recommendations with regard to such principles to the members of the United Nations.

The agenda of DISEC traditionally encompasses nuclear weapons and other weapons of mass destruction, the prevention of an arms race in outer space, conventional weapons, regional approaches to disarmament and security, confidence-building measures, and other initiatives aimed at reducing military tensions and strengthening the international security architecture.

AGENDA OVERVIEW

Outer Space has become an indispensable domain for the functioning of modern civilisation. Satellite systems provide critical services in communications, navigation and timing, along with helping us track natural disasters and meteorological anomalies. These services support both civilian and governmental activities and are the backbone of international peace and security.

As the dependence on space-based infrastructure has increased, so has the concern regarding the potential for interference with or denial of access to such systems during periods of tension or armed conflict on Earth. Anti-Satellite Weaponry (ASATs) and Directed Energy Weapons (DEWs) represent two correlated capabilities that particularly raise questions for international security. ASAT systems host a wide spectrum of means capable of disturbing, degrading or permanently destroying space infrastructure. These include kinetic systems such as co-orbital interceptors that achieve effect through physical impact and non-kinetic systems, including jamming or spoofing, along with directed energy applications.

The international community has long recognised that the prevention of an arms race in outer space is essential to the maintenance of international peace and security. The Treaty on Principles Governing the Activities of States in the Exploration and Use of Outer Space, including the Moon and Other Celestial Bodies (Outer Space Treaty, 1967) remains the cornerstone of the legal regime governing outer space activities. Article IV prohibits the placement in orbit of nuclear weapons or any other kinds of weapons of mass destruction. Article IX requires States Parties to conduct activities with due regard to the corresponding interests of all other States Parties and to avoid harmful contamination of outer space. The Treaty does not, however, expressly prohibit conventional ASAT systems or directed energy applications.

BACKGROUND

Key Concepts

Anti-Satellite Weaponry (ASATs)

Anti-Satellite weapons are designed to incapacitate or destroy space-based infrastructure. The UN classifies these into kinetic physical (direct-ascent missiles) and non-

kinetic (cyber and electronic means that disable systems without physical destruction.[1]



An Indian DRDO anti-satellite weapon (ASAT) in New Delhi in January 2020.

Source: Arms Control Association

Directed Energy Weaponry (DEWs)

Directed Energy Weaponry aims to create projects that use highly focused energy such as high-energy lasers or high-power microwaves to damage targets. Unlike conventional weaponry, DEWs can deliver effects at the speed of light and are highly scalable, ranging from dazzling lights to structural destruction.[2]

Non-kinetic Warfare

Non-Kinetic Warfare refers to military engagements that neutralise targets without conventional physical explosive force. This includes electronic warfare methods such as jamming and spoofing, or performing cyberattacks on enemy telemetry centres.[3]

Historical Context

The origins of ASAT Technology can be traced back to Cold War nuclear tests that highlighted the devastating effect of Electromagnetic Pulses (EMPs) on satellites.[4] Following the entry into force of the Partial Test Ban Treaty (1963) and the Outer Space Treaty (1967), which prohibited the placement of nuclear weapons and other weapons of mass destruction in orbit, research shifted toward conventional kinetic systems.[5][6] Both the United States and the Soviet Union pursued co-orbital ASAT programmes in which an interceptor satellite would manoeuvre into proximity with a target and destroy it through explosion or collision.[7]

The 21st century has seen multiple direct-ascent kinetic ASAT tests by major powers resulting in thousands of trackable debris fragments.[8] Parallel to kinetic developments, directed energy weapons (DEWs) evolved from the large, chemically powered laser systems researched during the Strategic Defence Initiative era of the 1980s to more compact, electrically powered solid-state lasers and high-power microwave

[1] A Lexicon for Outer Space Security. (n.d.-d). UNIDIR Lexicon entries on Kinetic Physical/Hard-Kill. https://spacesecuritylexicon.org/pdf/UNIDIR_A_Lexicon_for_Outer_Space_Security_en.pdf
[2] (2024, July 11). *Directed energy weapons: A new look at an "old" technology* . United Nations Institute for Disarmament Research. https://unidir.org/directed-energy-weapons-a-new-look-at-an-old-technology/
[3] UNITED NATIONS. (n.d.-u). Resolution 78/20: Reducing space threats through norms, rules and principles of responsible behaviours. https://digitallibrary.un.org/record/4030053?ln=en
[4] Effects of Electromagnetic Pulses . (2022a, November 10). https://www.aps.org/apsnews/2022/11/electromagnetic-pulse
[5] The Nuclear Threat Initiative. (n.d.-o). Partial Test Ban Treaty (PTBT). https://media.nti.org/documents/ptbt_partial_test_ban_treaty.pdf
[6] United Nations Treaty Collection. (n.d.-b). Outer Space Treaty (OST). https://treaties.un.org/doc/Publication/UNTS/volume%20610/volume-610-I-8843-English.pdf
[7] Papastergiou, S. (2024). Threats and vulnerability of satellite systems [Master thesis]. National and Kapodistrian University of Athens.
[8] Arms Control Association. (n.d.-a). Chinese Satellite Destruction . https://www.armscontrol.org/act/2007-03/chinese-satellite-destruction-stirs-debate

systems.[9] Contemporary DEW programmes emphasise mobility, integration with conventional air and missile defence architectures, and the ability to produce reversible (dazzling or temporary disruption) or irreversible effects against sensors and electronics.

KEY FRAMEWORKS

Outer Space Treaty (OST)

The Treaty on Principles Governing the Activities of States in the Exploration and the Use of Outer Space, including the Moon and Other Celestial Bodies, is commonly known as the Outer Space Treaty. The OST was opened for signature on 27 January 1967 and entered into force on 10 October 1967; it currently has over 110 State Parties, including all major spacefaring countries.[10]

The treaty established that the exploration and use of outer space shall be carried out for the benefit and in the interests of all countries and shall be in the province of all humankind. Article I. Outer Space, including the Moon and all other Celestial Bodies, is not subject to national appropriation by claim of sovereignty, by means of use or occupation, by any other means.[11]

Article II states that States Parties shall carry on activities in accordance with international law, including the Charter of the United Nations, in the interest of maintaining international peace and security and promoting international cooperation and understanding. Article III. Article IV retains more relevance towards the discussions on weaponry. It obligates States Parties not to place in orbit around the Earth any objects carrying nuclear weapons or any other kinds of weapons on celestial bodies, or station such weapons in outer space in any manner.

The treaty does not, however, explicitly prohibit conventional weapons or anti-satellite systems. Article IX requires States Parties to conduct all activities in outer space with due regard to the corresponding interests of all other States Parties and to avoid harmful contamination of outer space. The Outer Space Treaty remains the cornerstone against which all subsequent discussions on the prevention of an arms race in outer space, anti-satellite testing, and the long-term sustainability of space activities are measured.



The signing of the Outer Space Treaty in 1967.

Source: Space Foundation

Environmental Modification Convention (ENMOD)

Formally known as the Convention on the Prohibition of Military or Any Other Hostile Use of Environmental Modification Techniques, the Environmental Modification Convention (ENMOD) is a 1976 treaty established to prevent the use of environmental modification techniques having widespread, long-lasting or severe effects during times of conflict.

[9] U.S. Department of State. (n.d.-w). Strategic Defense Initiative. https://2001-2009.state.gov/r/pa/ho/time/rd/104253.htm
[10] AUDIOVISUAL LIBRARY OF INTERNATIONAL LAW. (n.d.-g). Treaty on principles governing the activities of states . https://legal.un.org/avl/ha/tos/tos.html
[11] *The Outer Space Treaty*. United Nations : Office for Outer Space Affairs. (n.d.). https://www.unoosa.org/unoosa/en/ourwork/spacelaw/treaties/outerspacetreaty.html

Article II of the treaty defines 'environmental modification' as "any technique for changing - through the deliberate manipulation of natural processes - the dynamics, composition or structure of the earth, including its biota, lithosphere, hydrosphere and atmosphere, or of outer space". The convention promotes collaboration by mandating scientific and technological information exchange between member nations, while remaining cognizant of peaceful uses of environmental modification. While focusing primarily on the military uses of environmental modification, the convention also covers non-military environmental changes.[12]

The ENMOD does not directly ban the use of DEWs. However, if a DEW is used in a particularly environmentally harmful manner, the ENMOD can be triggered to tackle it. For instance, if the use of a high-energy laser beam is known to alter weather patterns or disturb local ecosystems, its deployment is prohibited under the ENMOD.[13]

Critics of the ENMOD have argued that the requirements for environmental modification techniques to be "widespread, long-lasting or severe" in order to be prohibited allow localised and short-term uses of environmental warfare to go unchecked.[14]

UNGA Resolution 77/41

United Nations General Assembly resolution 77/41, entitled "Destructive direct-ascent anti-satellite missile testing," was adopted on 7 December 2022 by a recorded vote of 155 in favour, 9 against and 9 abstentions. It represents the most specific multilateral political commitment to date addressing a particular category of anti-satellite testing.

The resolution expresses serious concern about the possibility of an arms race in outer space and reaffirms that the prevention of such an arms race is in the interest of maintaining international peace and security. It further expresses concern that the use of destructive anti-satellite systems might have widespread and irreversible impacts on the outer space environment. In its operative paragraphs, the General Assembly: "Calls upon all States to commit not to conduct destructive direct-ascent anti-satellite missile tests; Considers such a commitment to be an urgent, initial measure aimed at preventing damage to the outer space environment, while also contributing to the development of further measures for the prevention of an arms race in outer space; Calls upon all States to continue discussions in the relevant bodies and to establish and develop further practical steps that could be taken in order to enable risk reduction, prevent conflict from occurring in outer space and prevent an arms race in outer space".[15]

Although non-binding, resolution 77/41 has contributed to the development of a political norm against debris-generating direct-ascent ASAT tests.

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[12] *CONVENTION ON THE PROHIBITION OF MILITARY OR OTHER HOSTILE USE OF ENVIRONMENTAL MODIFICATION TECHNIQUES* . United Nations. (n.d.-a). https://treaties.un.org/doc/Treaties/1978/10/19781005%2000-39%20AM/Ch_XXVI_01p.pdf

[13] *ICRC's Legal Factsheet Insert Image here 1976 convention on the prohibition of*. International Committee of the Red Cross. (n.d.). https://www.icrc.org/sites/default/files/2026-05/1976-ENMOD-ICRC-factsheet.pdf

[14] DIKGALE, E. M. (2024). *Challenges in regulating environmental modification techniques in an armed conflict : An analysis of the ENMOD convention - proquest*. ProQuest. https://www.proquest.com/openview/fbfb4523f78806d5e3dff76bc4d17983/1?pq-origsite=gscholar&cbl=2026366&diss=y

[15] *A/RES/77/41 - general assembly - the united nations*. UNITED NATIONS. (n.d.). https://documents.un.org/doc/undoc/gen/n22/738/92/pdf/n2273892.pdf

Prevention of an Arms Race in Outer Space (PAROS)

Prevention of an Arms Race in Outer Space (PAROS) is both a long-standing agenda item of the United Nations General Assembly and the Conference on Disarmament, and a broader policy objective of the international community.[16] The concept was formally introduced in the Final Document of the Tenth Special Session of the General Assembly devoted to disarmament (1978), which stated that further measures should be taken and appropriate international negotiations held, in accordance with the spirit of the Outer Space Treaty, in order to prevent an arms race in outer space.[17]

Since 1981, the General Assembly has annually adopted resolutions under the item "Prevention of an arms race in outer space." These resolutions reaffirm that the exploration and use of outer space shall be for peaceful purposes, call upon all States, particularly those with major space capabilities, to contribute actively to the objective of the peaceful use of outer space and the prevention of an arms race, and reiterate that negotiations for the conclusion of an international agreement or agreements to prevent an arms race in outer space remain a priority task of the Conference on Disarmament.

The Conference on Disarmament established an Ad Hoc Committee on PAROS in 1985, which met until 1994. Subsequent efforts have included subsidiary bodies within the Conference, Groups of Governmental Experts (notably in 2018–2019 and more recently), and Open-Ended Working Groups examining both further practical measures for PAROS and the reduction of space threats through norms, rules and principles of responsible behaviours.

CHALLENGES

Lack of Binding Global Regulatory Mechanisms

There is no existing international treaty banning the use of ASATs and DEWs as a class of weapons. The bedrock of space law is the 1967 Outer Space Treaty (OST), which was crafted against the backdrop of the Cold War's nuclear threats in outer space. While OST Article IV prevents the placement of nuclear weapons and other weapons of mass destruction in orbit, there are no prohibitions regarding conventional weapons or the general military exploitation of outer space. Hence, the launch of missiles from the ground to destroy satellites, killer satellites co-orbiting with other satellites, and lasers blinding satellite sensors all lie in a grey area.[18]

Efforts at closing this gap have been repeatedly stalled. First, the Conference on Disarmament, which is mainly responsible for negotiations on the Prevention of an Arms Race in Outer Space (PAROS), has been deadlocked and unable to engage in any substantive negotiations for more than two decades.[19] Second, legally binding initiatives such as the Treaty on the Prevention of the Placement of Weapons in Outer Space (PPWT), as well as politically binding initiatives such as the International Code of Conduct for Outer Space Activities, have never reached formal negotiations. The PPWT has never been formally negotiated at the Conference on

[16] United Nations General Assembly, "Prevention of an arms race in outer space," resolution adopted at the 53rd session, 1998. https://documents.un.org/doc/undoc/gen/n98/361/31/pdf/n9836131.pdf;
[17] United Nations Dag Hammarskjöld Library, "General Assembly & Disarmament" research guide. https://research.un.org/en/docs/disarmament/intro
[18] Global Public International Law (University of Bonn), "Preventing an arms race in outer space and political game-play at the United Nations." https://gpil.jura.uni-bonn.de/?p=2150
[19] Project Ploughshares, "Statement to the 74th Session of the General Assembly First Committee," Oct. 2019. https://ploughshares.ca/statement-to-the-74th-session-of-the-general-assembly-first-committee/

Disarmament, and talks on the Code, which were due to begin at the UN in July 2015, stalled when states could not agree on its scope or on which UN body should negotiate it.[20] [21]

The PPWT was proposed jointly by Russia and China in 2008 and amended in 2014,[22] yet Western states have refused to accept it. Germany, for instance, has said the draft is not a sufficient basis for a treaty,[23] while the United States has criticised it for lacking any verification mechanism and for leaving ground-launched ASAT missiles unrestricted.[24]

It is clear from such recurring impasses that neither non-binding resolutions nor voluntary commitments alone can prevent the evolution of ASATs and DEWs. What is needed to bridge the loopholes in the Outer Space Treaty is a binding, verifiable regime regulating both conventional and non-kinetic weapons.

Asymmetric Nature of DEWs

Directed energy weapons present problems that conventional arms control was not designed to handle. The only binding international agreement that touches on them is narrow. Protocol IV to the Convention on Certain Conventional Weapons prohibits laser weapons specifically designed, as their sole combat function or as one of their combat functions, to cause permanent blindness to unenhanced human vision. However, it explicitly excludes blinding that occurs as an incidental or collateral effect of the legitimate military use of laser systems, including lasers used against optical equipment.[25]

Dazzlers that only temporarily impair vision are not prohibited, although parties must take feasible precautions to avoid incidental blinding, and the protocol binds only those states that have separately consented to it.[26] Lasers aimed at satellite sensors, drones, or missiles therefore remain essentially unregulated.

The first aspect of asymmetry pertaining to DEWs is the difficulty of attributing DEW attacks to specific actors. When a laser dazzles a satellite, it may be possible to locate the laser at the moment of the attack, but because such lasers can be mobile, attributing the attack to a particular actor becomes difficult; the attacker does not need to be on its own territory, or even on the same continent.[27]

Because laser attacks produce no debris and their source is hard to trace, the general public may never learn that an attack took place unless the satellite operator chooses to disclose it.[28] This allows states to operate in a 'grey zone' below the threshold of open conflict, weakening deterrence and making violations difficult to prove.

[20] Project Ploughshares, "Statement to the 74th Session of the General Assembly First Committee," Oct. 2019. https://ploughshares.ca/statement-to-the-74th-session-of-the-general-assembly-first-committee/
[21] T. Farnsworth, "No End in Sight for Space Code," *Arms Control Today*, Sept. 2015. https://www.armscontrol.org/act/2015-09/news/no-end-sight-space-code
[22] Permanent Mission of Pakistan to the UN, Geneva, "Statement on Thematic Debate on Outer Space (Disarmament Aspects)." https://pakungeneva.pk/TopicDetail.aspx?ID=1270
[23] Global Public International Law (University of Bonn), "Preventing an arms race in outer space and political game-play at the United Nations." https://gpil.jura.uni-bonn.de/?p=2150
[24] SpaceNews, "U.S. Review of Russian-Chinese Space Weapons Treaty Finds Same Flaws," Sept. 2014. https://spacenews.com/?p=15820
[25] Protocol on Blinding Laser Weapons (Protocol IV to the CCW), 13 Oct. 1995, text via Federation of American Scientists. https://nuke.fas.org/control/ccw/text/protocol4.htm
[26] Model Diplomat, "Protocol IV (CCW)." https://modeldiplomat.com/learn/glossary/protocol-iv-ccw
[27] CSIS Aerospace Security Project, "Counterspace Weapons 101." https://aerospace.csis.org/aerospace101/counterspace-weapons-101
[28] Air & Space Forces Magazine, "'Gray Zone' Conflict Drives Space Weapons Development." https://airandspaceforces.com/?p=65354

The second aspect of asymmetry pertaining to DEWs is the very low cost of using them compared with conventional weapons. Because a laser is fired using electrical power rather than physical ammunition, it offers a far lower cost per shot and an almost unlimited number of shots compared with the missile interceptors traditionally used against the same targets.

When Israel tested its Iron Beam laser in 2022, officials put the energy cost of a single shot at $3.50. Once maintenance and depreciation are included, estimates rise to about $2,000 per interception, which is still considerably cheaper than an air defence missile.[29] In December 2025, Israel's Ministry of Defence delivered Iron Beam to the Israeli Defence Force, making it the world's first operational high-energy laser interceptor. [30] Near-zero firing costs could lower the threshold for using force, and they give an enormous advantage to states that can afford the high upfront costs of research and infrastructure, widening the technological gap between them and everyone else.

The third aspect of asymmetry pertaining to DEWs is that they can be developed and acquired with far less visibility than missile-based weapons. Laser weapons are developed and tested at ground facilities rather than through visible launches, so other states often know little about how many exist or how capable they are. Russia has a mobile laser dazzling system that may be intended to target the optical sensors of imaging satellites, while China is likely developing directed energy weapons for counterspace use, with at least five laser testing facilities, although little is publicly known about any fielded capabilities.[31]

Because laser development leaves far fewer observable traces than missile testing, confidence-building measures and verification mechanisms are much harder to design.

Threat of Conflict in Outer Space

The use of destructive ASATs threatens not only the target but also every state that relies on space infrastructure. In November 2021, Russia destroyed its own COSMOS 1408 satellite in a direct-ascent ASAT test, generating more than 1,500 pieces of trackable debris, which was expected to produce hundreds of thousands of smaller fragments.[32] The debris immediately threatened the International Space Station, forcing its seven astronauts to shelter in docked capsules. Similarly, China's 2007 ASAT test created a field of around 3,500 pieces of debris, most of which is still in orbit.[33] As of early 2025, counterspace testing by the US, Russia, China, and India had generated 6,851 catalogued pieces of trackable debris, of which 2,920 remained in orbit.[34] Because debris does not discriminate between states, a single conflict could trigger cascading collisions that leave key orbits unusable for decades.

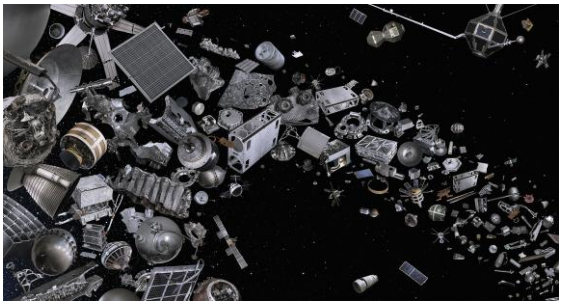
The proliferation of counterspace technologies is also a growing threat. According to the Secure World Foundation's 2025 assessment, 12 countries are developing counterspace capabilities, although only non-destructive

[29] Mezha.media, "Iron Beam: new Israeli laser-based air defense system," Apr. 2022. https://mezha.media/en/2022/04/15/iron-bea-new-israeli-laser-based-air-defense-system
[30] Jewish Virtual Library "Or Eitan (Iron Beam) Laser Defense System."
[31] NewSpace Economy, "The Evolving Landscape of Global Counterspace Capabilities" (summary of SWF 2024 report), May 2024. https://newspaceeconomy.ca/2024/05/10/the-evolving-landscape-of-global-counterspace-capabilities/
[32] U.S. Office of Space Commerce, "U.S. Response to Russian Anti-Satellite Test." https://space.commerce.gov/u-s-response-to-russian-anti-satellite-test/
[33] The Heritage Foundation, "Russian ASAT-Test Debris Threatens International Space Station and Escalates Global Tensions," Nov. 2021. https://www.heritage.org/sites/default/files/2021-11/FS224.pdf
[34] Resilient Navigation and Timing Foundation, "Counterspace – Annual report from Secure World Foundation," Apr. 2025. https://rntfnd.org/2025/04/06/counterspace-annual-report-from-secure-world-foundation/

capabilities are currently being used in active military conflicts.[35]

The deployment of co-orbital anti-satellite systems is also a growing threat. Co-orbital systems are satellites placed in orbit that can be manoeuvred towards a target satellite in order to attack it, for example by colliding with it or by detonating an explosive nearby. In May 2024, Russia launched COSMOS 2576 into the same orbital ring as USA 314, a US reconnaissance satellite, where it appeared to be trailing USA 314's path at a faster speed, suggesting the two would eventually come into closer proximity.[36] Close approaches like this can be read as preparation for an attack, raising the risk of miscalculation.

The gravest concern is the possibility of nuclear weapons in orbit. In April 2024, Russia vetoed a US-Japan Security Council resolution calling on states to prevent an arms race in outer space (13 in favour, China abstaining), amid US claims that Russia is developing a nuclear-armed satellite.[37] US officials have said no such weapon has been deployed.[38] A Russia-China amendment to ban all weapons in space, not just WMDs, failed 7-7.[39] Both sides claimed to oppose weaponising space, yet neither accepted the other's framework.



Space Debris Around Earth

Source: European Space Agency

IMPLICATIONS

Civilian Infrastructure

Attacks against satellites would damage the country that owns the targeted satellite and also the civilians who rely on it. The International Committee of the Red Cross (ICRC) warns that kinetic or non-kinetic attacks in outer space could have significant effects on civilians on Earth, because technology enabled by space systems is used in most areas of civilian life.[40]

Many satellites are "dual-use", serving both civilian and military purposes, and the ICRC notes that this could make them military objectives in certain circumstances. [41] Such attacks have already affected civilians far from any battlefield.

[35] Secure World Foundation, "2025 Global Counterspace Capabilities Report." https://swfound.org/publications-and-reports/2025-global-counterspace-capabilities-report
[36] Al Jazeera, "Russia launches 'space weapon' in path of US satellite: Pentagon," May 2024. https://www.aljazeera.com/news/2024/5/22/russia-launches-space-weapon-in-path-of-us-satellite-pentagon
[37] The Star (Kenya), "Russia vetoes UN vote on stopping arms race in outer space," Apr. 2024. https://www.the-star.co.ke/news/2024-04-25-russia-vetoes-un-vote-on-stopping-arms-race-in-outer-space
[38] Al Jazeera, "Russia launches 'space weapon' in path of US satellite: Pentagon," May 2024. https://www.aljazeera.com/news/2024/5/22/russia-launches-space-weapon-in-path-of-us-satellite-pentagon
[39] Space.com, "Russia vetoes UN resolution against nuclear weapons in space," Apr. 2024. https://www.space.com/russia-nuclear-weapons-space-veto-un-resolution
[40] International Committee of the Red Cross, "The potential human cost of the use of weapons in outer space and the protection afforded by international humanitarian law," position paper, 8 Apr. 2021. https://international-review.icrc.org/articles/the-potential-human-cost-weapons-in-outer-space-and-protection-afforded-by-ihl-icrc-position-paper-915
[41] International Committee of the Red Cross, "The potential human cost of the use of weapons in outer space and the protection afforded by international humanitarian law," position paper, 8 Apr. 2021. https://international-review.icrc.org/articles/the-potential-human-cost-weapons-in-outer-space-and-protection-afforded-by-ihl-icrc-position-paper-915

On 24 February 2022, the day Russia invaded Ukraine, a cyberattack on Viasat's KA-SAT network disabled satellite modems in Ukraine and cut off remote monitoring and control of 5,800 wind turbines in Germany.[42] Later in 2022, a senior official of the Russian Foreign Ministry told a UN General Assembly meeting that commercial satellites helping Ukraine could become legitimate targets for retaliation.[43] As a single satellite increasingly serves both military and civilian users, a binding mechanism is needed to define how such satellites may be treated during conflict.

Critical Infrastructure

Satellites are also important for critical infrastructure such as power grids, banking systems and communication networks, which rely on satellite navigation signals for precise time synchronisation.[44] A 2019 study commissioned by the US National Institute of Standards and Technology estimated that a GPS outage could cost the US economy about USD 1 billion per day.[45] A nuclear explosion in space would cause far greater damage.

In 1962, the US Starfish Prime test blew out several hundred streetlights in Hawaii, around 1,500 kilometres away, and damaged at least six satellites, all of which eventually failed. [46] In 2024, a senior Pentagon official warned that a sufficiently powerful detonation in the right location could render low Earth orbit unusable for up to a year. [47]As much of the modern economy relies on a few satellite systems, safeguarding satellites that provide critical services might become a necessity in any future agreement.

Environmental Degradation

The use of destructive ASATs pollutes orbital space, which is a limited resource shared by all countries. The European Space Agency (ESA) estimates that over 1.2 million pieces of debris larger than 1 centimetre, each capable of causing catastrophic damage, are currently in orbit.[48]

ESA also warns that debris would keep accumulating even if no new satellites were launched, a runaway chain reaction known as Kessler syndrome that could render certain orbits unusable.[49]

The altitude at which an ASAT test is carried out largely determines how long its debris lasts. While China's 2007 test at 865 kilometres produced about 3,000 pieces of debris that will remain for decades, most of the fragments from

[42] SentinelOne Labs, "AcidRain | A Modem Wiper Rains Down on Europe," 2022. https://www.sentinelone.com/labs/acidrain-a-modem-wiper-rains-down-on-europe/
[43] Anadolu Agency, "US vows to respond if Russia attacks US civilian satellites," 27 Oct. 2022. https://aa.com.tr/en/americas/us-vows-to-respond-if-russia-attacks-us-civilian-satellites/2722698
[44] International Committee of the Red Cross, "The potential human cost of the use of weapons in outer space and the protection afforded by international humanitarian law," position paper, 8 Apr. 2021. https://international-review.icrc.org/articles/the-potential-human-cost-weapons-in-outer-space-and-protection-afforded-by-ihl-icrc-position-paper-915
[45] U.S. Office of Space Commerce, "DOC Study on Economic Benefits of GPS," June 2019. https://space.commerce.gov/?p=3201
[46] P. Plait, "The 50th anniversary of Starfish Prime: the nuke that shook the world," SYFY Wire, 9 July 2012. https://www.syfy.com/syfy-wire/the-50th-anniversary-of-starfish-prime-the-nuke-that-shook-the-world
[47] Air & Space Forces Magazine, "DOD Official Confirms Russia Is Developing an 'Indiscriminate' Space Nuke," 2024. https://www.airandspaceforces.com/dod-official-russia-indiscriminate-space-nuke/
[48] European Space Agency, "ESA Space Environment Report 2025," Apr. 2025. https://www.esa.int/Space_Safety/Space_Debris/ESA_Space_Environment_Report_2025
[49] UN-SPIDER, "ESA Space Environment Report 2025." https://un-spider.org/news-and-events/news/esa-space-environment-report-2025

India's 2019 test at 282 kilometres were expected to decay within weeks or months.[50]

In spite of these problems, the Space Debris Mitigation Guidelines adopted by the Inter-Agency Space Debris Coordination Committee in 2002 remain purely voluntary.[51] Since debris created by one country threatens the satellites of all countries, voluntary guidelines alone may not be sufficient, and binding rules might become a necessity.

Risk of Permanent Injury to Victims

Lasers pose a direct danger of permanent injury, since the human eye focuses laser light onto the retina and can concentrate its energy up to 100,000 times.[52] As retinal tissue cannot regenerate, such an injury is permanent.[53]

Military-grade lasers have already injured people outside combat zones. In 2018, the United States said that lasers from the vicinity of the Chinese base in Djibouti had caused minor eye injuries to two American pilots, an allegation China denied.[54] In 2023, the Philippine Coast Guard stated that a China Coast Guard laser had temporarily blinded its crew. [55] However, Protocol IV of the CCW only prohibits lasers specifically designed to cause permanent blindness, and excludes incidental blinding from its scope.[56]

With directed energy weapons such as Israel's Iron Beam now in operational service, the committee must consider regulating the actual effects of such weapons on people, and not only the purpose for which they were designed.[57]

CASE STUDIES

Russia-Ukraine War

After Ukrainian President Viktor Yanukovych rejected a deal for greater economic integration with the European Union (EU), riots broke out in Ukraine.[58] Under the pretext of protecting Russian citizens, Moscow sent its military to Crimea.[59] On February 24, 2022, Moscow declared a full-scale invasion to 'demilitarise and denazify' the country. [60]

During this war, Directed Energy Weapons (DEWs) and anti-satellite (ASAT) weapons appear as emerging threats, not as the primary method of conflict. Drones currently define the

[50] Carnegie Endowment for International Peace, "India's ASAT Test: An Incomplete Success," 15 Apr. 2019. https://carnegieendowment.org/research/2019/04/indias-asat-test-an-incomplete-success
[51] UN-SPIDER, "ESA Space Environment Report 2025." https://un-spider.org/news-and-events/news/esa-space-environment-report-2025
[52] Princeton University Environmental Health and Safety, "Section 2: Laser Hazards." https://ehs.princeton.edu/book/export/html/363
[53] Wake Forest University Environmental Health and Safety, "Laser Safety Manual." https://prod.wp.cdn.aws.wfu.edu/sites/208/2014/08/ehs-laser-safety-manual.pdf
[54] Defense News, "China denies using lasers on US aircraft in Djibouti," 6 May 2018. https://www.defensenews.com/flashpoints/2018/05/06/china-denies-using-lasers-on-us-aircraft-in-djibouti/
[55] Philippine Daily Inquirer, "PCG: Crew affected by Chinese laser attack being monitored for adverse effects," 14 Feb. 2023. https://newsinfo.inquirer.net/?p=1729763
[56] Protocol on Blinding Laser Weapons (Protocol IV to the CCW), 13 Oct. 1995, text via Federation of American Scientists. https://nuke.fas.org/control/ccw/text/protocol4.htm
[57] Jewish Virtual Library, "Or Eitan (Iron Beam) Laser Defense System." https://www.jewishvirtuallibrary.org/iron-beam-laser-defense-system
[58] Council on Foreign Relations). War in ukraine | global conflict tracker. Council on Foreign Relations. https://www.cfr.org/global-conflict-tracker/conflict/conflict-ukraine
[59] Transcript: Putin says Russia will protect the rights of Russians abroad - The Washington Post. https://www.washingtonpost.com/world/transcript-putin-says-russia-will-protect-the-rights-of-russians-abroad/2014/03/18/432a1e60-ae99-11e3-a49e-76adc9210f19_story.html
[60] Kirby, P. (2026, February 24). Why did Putin's Russia invade Ukraine and how could the war end? BBC News. https://www.bbc.com/news/articles/cj0q964851po

fighting, given Ukraine has invested millions in building roughly 2 million drones in 2024, with both sides using jamming to interfere with operations.[61]

Recently, however, DEWs are being studied as a counter-drone option, with high-energy lasers damaging targets alongside microwaves frying their electronics. The UK's DragonFire reportedly costs about 10(pound sign) per unit, making it an extremely attractive option. However, RAND advises against using the technology as its research is in its infancy. A Ukrainian laser called Tryzub was announced, but its claims remain unverified.[62]

Russia has fielded ground-based lasers and ASAT systems to interfere with Western intelligence systems. In November 2021, an ASAT test destroyed one of their own satellites and created over 1,500 pieces of debris.[63] That test has been interpreted as an attempt to intimidate by showcasing the technology. Russia has since been using spoofing and jamming as methods of interference, as displayed in the 2022 Viasat attack, with real use of the technology currently kept on hold.[64]

ASATs have not been used offensively by either side, as fears of debris may ensue global controversy or intervention. Risks of ASATs were clearly displayed in the 2021 test, and could be taken as a show of power rather than actual use.[65] Russia is unlikely to use ASATs unless it perceives an existential threat. Currently, Ukraine's biggest space-related threat remains GPS jamming and spoofing.[66] However, in 2024, a concern grew over a Russian armed ASAT in development, which violates the Outer Space Treaty. It is perceived as a coercive tool rather than an actual threat due to its impractical nature.[67]



A rescuer worker walks through the rubble damaged in a Russian ballistic missile strike in Kyiv.

Source: CNN

[61] Bondar, K. (2025, March 6). Ukraine's future vision and current capabilities for waging AI-enabled autonomous warfare. Center for Strategic and International Studies. https://www.csis.org/analysis/ukraines-future-vision-and-current-capabilities-waging-ai-enabled-autonomous-warfare

[62] Dee et al.(2025). Invisible defender? Opportunities and challenges for integrating DEWs into Ukraine's C-UAS framework (RRA3833-7). RAND Corporation. https://www.rand.org/content/dam/rand/pubs/research_reports/RRA3800/RRA3833-7/RAND_RRA3833-7.pdf

[63] Sankaran, J. (2022, March). Russia's anti-satellite weapons: An asymmetric response to U.S. aerospace superiority. Arms Control Today. https://www.armscontrol.org/act/2022-03/features/russias-anti-satellite-weapons-asymmetric-response-us-aerospace-superiority

[64] Ogden et al (2024, February). The role of the space domain in the Russia-Ukraine war and the impact of converging space and AI technologies (CETaS Expert Analysis). The Alan Turing Institute. https://cetas.turing.ac.uk/publications/role-space-domain-russia-ukraine-war

[65] House of Lords International Relations and Defence Committee. (2024). UK Parliament. https://publications.parliament.uk/pa/ld5901/ldselect/ldintrel/10/1008.htm

[66] Mulder, C. P. (2022, March 26). Russia's space weapons may be the next frontier in the Ukraine conflict. Atlantic Council. https://www.atlanticcouncil.org/blogs/new-atlanticist/russias-space-weapons-may-be-the-next-frontier-in-the-ukraine-conflict/

[67] Davis, M. (2024, March 7). Russian anti-satellites: The re-emerging threat of orbital nuclear weapons? King's College London. https://www.kcl.ac.uk/the-re-emerging-threat-of-orbital-nuclear-weapons

Israel-Lebanon Conflict

Israel invaded Lebanon in 1982, responding to the Palestine Liberation Organisation attacks, leading to Hezbollah being created as a retaliation. Further wars followed in 1993, 1996 and 2006. After Hamas' October 2023 attack, Hezbollah began firing on Israel in support of Gaza, with Israel fighting back.[68] Hezbollah is represented in Lebanon's governments and controls the country's most powerful armed force. It is deemed to be a terrorist organisation by Israel, the US and the UK. The conflict escalated in September 2024 with exploding pagers, the killing of leader Hassan Nasrallah and then the ground assault in southern Lebanon on 1st October.[69]

The Israel-Lebanon conflict produced the first publicly acknowledged combat use of Directed Energy Weapons (DEWs). Hezbollah fired more than 300 explosive drones at Israel after 8th October, 2023, many of which evaded air defences. In May 2025, the IDF revealed its air force had intercepted the drones with a low-powered version of the Iron Beam, reporting high interception rates.[70] The ministry claims to have deflected waves of the drones, acknowledging first-hand use of the directed energy weapons.[71]

Although officials did not state the number of interceptions, each shot reportedly costs under 5$, as compared to the tens of thousands of dollars used for an Iron Dome missile. However, the technology does suffer from poor performance against evasive targets and during bad weather.[72] The Iron Beam can engage rockets, drones and cruise missiles at up to 10 km, with the downside of the weapon needing to cool off after overheating very often.[73] Lasers were reportedly used again in March 2026 against Hezbollah rockets, though the performance of the Iron Beam remains unclear. It is a ground-based and an orbital laser, although while able to target satellites, it suits short-range missiles poorly.[74]

Defence Minister Katz now wants Israel to lead in space-based lasers, a leap beyond the Iron Beam. Orbit raises problems of power and accuracy, however. The claim that no country has space attack capability is doubtful, as Russia and China have successfully conducted ASAT tests, which raised debris concerns.[75]

CONCLUSION

The growing capabilities of anti-satellite systems and directed energy weapons present complex challenges to the maintenance of global peace

[68] Al Jazeera Staff. (2024, September 18). The history of conflict between Hezbollah and Israel. Al Jazeera. https://www.aljazeera.com/news/2024/9/18/hezbollah-and-israel-a-timeline-of-conflict
[69] Taylor, I. (2024, October 6). Two weeks of turmoil: A timeline of Israel and Hezbollah. Next Century Foundation. https://www.nextcenturyfoundation.org/two-weeks-of-turmoil-a-timeline-of-israel-and-hezbollah/
[70] Fabian, E. (2025, May 28). IDF reveals it used laser system to intercept dozens of Hezbollah drones last year. The Times of Israel. https://www.timesofisrael.com/idf-reveals-it-used-laser-system-to-intercept-dozens-of-hezbollah-drones-last-year/
[71] Israel acknowledges use of laser weapons to intercept drones. (2025, May 28). Bloomberg. https://www.bloomberg.com/news/articles/2025-05-28/israel-acknowledges-use-of-laser-weapons-to-intercept-drones
[72] Robertson, R., & Jackson, W. (2025, May 30). Israel confirms first use of laser weapons in combat. Straight Arrow News. https://san.com/cc/israel-confirms-first-use-of-laser-weapons-in-combat/
[73] VisionIAS. (2025, June 2). Israel becomes first country to shoot down enemy drones with laser weapon. Vision IAS. https://visionias.in/current-affairs/news-today/2025-06-02/science-and-technology/israel-becomes-first-country-to-shoot-down-enemy-drones-with-laser-weapon
[74] Suciu, P. (2026, March 4). Israel's "Jewish space lasers" strike again—sort of. The National Interest. https://nationalinterest.org/blog/buzz/israels-jewish-space-lasers-strike-again-sort-of-ps-030426
[75] Katz claims that Israel is developing space lasers to lead the new technological war. (29 June, 2026). Aurora. https://www.aurora-israel.co.il/en/laseres-espaciales-de-israel-katz-plan-militar/

and security, along with preservation for long-term outer space activities. The potential for kinetic operations to generate persistent debris, together with the dual-use nature of non-kinetic means, underlines the need for careful consideration of both the legal and operational implications of such technologies. Existing means, notably the Outer Space Treaty and the ongoing multilateral processes such as the discussion on the Prevention of an Arms Race in Outer Space, provide the essential foundation. In addressing this agenda item, the Committee has an opportunity to reaffirm the principle that the exploration and use of outer space shall be carried out for the benefit and in the interests of all countries, and to encourage practical steps that promote the safety, security and sustainability of outer space activities.

Agenda 2 - Reviewing the Convention on Certain Conventional Weapons amidst an Evolving Military Landscape

AGENDA OVERVIEW

The Convention on Prohibitions or Restrictions on the Use of Certain Conventional Weapons Which May Be Deemed to Be Excessively Injurious or to Have Indiscriminate Effects (CCW), also known as the Inhumane Weapons Convention, constitutes one of the principal instruments of international humanitarian law regulating the means and methods of war.

Adopted in 1983, the CCW is a framework convention designed to prohibit or restrict the use of specific categories of conventional weapons that cause excessive injury or unnecessary suffering or that are inherently indiscriminate in their effects. The Convention operates through a series of annexed protocols, each of which addresses a particular weapon category.

Protocol I is on non-detectable fragments; Protocol II (and its 1996 Amended version) on mines, booby-traps and other devices; Protocol III on incendiary weapons; Protocol IV on blinding laser weapons (adopted in 1995 as a pre-emptive prohibition); and Protocol V on explosive remnants of war (adopted in 2003). This modular structure has enabled the CCW to adapt to the ever-changing core principles of international humanitarian law, particularly the principles of distinction, proportionality and the prohibition of unnecessary suffering. Rapid technological advances, notably in artificial intelligence, autonomy, and precision-guided munitions, have introduced new categories of weapons systems that were not contemplated when the Convention was drafted.

Recent withdrawals by certain States from related treaties, such as the Anti-Personnel Mine Ban Convention and the Convention on Cluster Munitions, have further underscored the tensions between humanitarian imperatives and perceived national security requirements.

The Seventh Review Conference of the CCW, scheduled for November 2026, will consider the future direction of this work, including the possibility of launching formal negotiations.

BACKGROUND

Key Definitions

Incendiary Weapons

Under Protocol III to the Convention on Certain Conventional Weapons, an "incendiary weapon" is defined as any weapon or munition which is primarily designed to set fire to an object or to cause burn injury to persons through the action of flame, heat or a combination thereof, produced by a chemical reaction of a substance delivered on the target. The definition explicitly excludes:

(a) Munitions which have incidental incendiary effects (such as illumination, tracers, smoke or signalling systems); and

(b) Munitions designed to combine penetration, blast or fragmentation effects with an additional incidental incendiary effect (such as armour-piercing projectiles or explosive bombs), where the incendiary effect is not specifically designed to cause burn injury to persons.

This design-based definition has been the subject of ongoing discussion, as certain multipurpose munitions (notably those containing white phosphorus) may produce severe burn effects

while falling outside the formal scope of the Protocol.[76]

Lethal Autonomous Weapons Systems (LAWS)

Within the framework of the Group of Governmental Experts on Emerging Technologies in the Area of Lethal Autonomous Weapons Systems (GCE on LAWS) under the CCW, a lethal autonomous weapon is generally characterised as a combination of one or more weapons and functionally integrated technological components that can identify, select and engage a target without intervention by a human operating the system. Key elements of the emerging characterisation include:

(a) The capacity to perform critical functions (identification, selection and engagement of targets) without human intervention in the execution of those tasks.

(b) The characterisation does not exclude systems that may also be used in non-lethal ways, nor does it prejudice future refinements.[77]

Historical Context

The Convention on Certain Conventional Weapons emerged from the growing global concern in the 1970s over the humanitarian impact of certain conventional weapons used in armed conflicts, particularly those that cause superfluous injury or have indiscriminate effects. Building on the earlier efforts within the United Nations and the Diplomatic Conference on the Reaffirmation and Development of International Humanitarian Law applicable in Armed Conflicts (1974-1977), the United Nations Conference on Prohibitions or Restrictions on the Use of Certain Conventional Weapons was convened in Geneva in 1979-1980.[78]

The Convention and its first three Protocols were adopted on 10 October 1980 and opened for signature on 10 April 1981, entering into force on 2 December 1983.[79] Subsequent Review Conferences and Meetings of High Contracting Parties have enabled the progressive development of the legal framework: Protocol IV on Blinding Laser Weapons was adopted in 1995; Protocol II was substantially amended in 1996; and Protocol V on Explosive Remnants of War was adopted in 2003. An amendment to Article 1 of the Convention in 2001 extended its application to non-international armed conflicts.

KEY FRAMEWORKS

Convention on Certain Conventional Weapons (CCW)

Protocol I

Non-Detectable Fragments (10 October 1980): Protocol I prohibits the use of any weapon where the primary effect of which is to injure by fragments which, in the human body, escape detection by X-rays. The Protocol aims to prevent

[76] International Humanitarian Law Databases. (n.d.-m). Definition of Incendiary Weapons. https://ihl-databases.icrc.org/en/ihl-treaties/ccw-protocol-iii-1980/article-1

[77] United Nations Office for Disarmament Affairs. (n.d.-w). Group of Governmental Experts on Emerging Technologies in the Area of Lethal Autonomous Weapons Systems. https://docs-library.unoda.org/Convention_on_Certain_Conventional_Weapons_-Group_of_Governmental_Experts_on_Lethal_Autonomous_Weapons_Systems_(2026)/CCW-GGE.1-2026-3-Advance_copy.pdf

[78] International Humanitarian Law Databases. (n.d.-p). Origins in the 1970s and concern over superfluous injury. https://ihl-databases.icrc.org/en/ihl-treaties/geneva-finact-1977

[79] International Humanitarian Law Databases. (n.d.-n). Adoption and entry into force of the Convention and first three Protocols. https://ihl-databases.icrc.org/es/ihl-treaties/ccw-1980

injuries that are particularly difficult to treat surgically.

Protocol II

Mines, Booby-Traps and Other Devices (10 October 1980):
The original Protocol II imposed restrictions on the use of mines, booby-traps and other devices. The Amended Protocol II (adopted 3 May 1996) significantly strengthened these provisions. It prohibits the use of non–detectable anti-personnel mines and their transfer, requires that remotely delivered mines be equipped with self-destruction and self-deactivation mechanisms, and mandates that mines used outside marked and monitored areas meet strict technical standards. Amended Protocol II is also the only legally binding instrument under the CCW that explicitly addresses improvised explosive devices (IEDs).

Protocol III

Incendiary Weapons (10 October 1980):
Protocol III regulates the use of incendiary weapons. It prohibits, in all circumstances, making the civilian population the object of attack by incendiary weapons. It further restricts the use of air-delivered incendiary weapons against military objectives located within concentrations of civilians. Discussions continue regarding the adequacy of its definition and the distinction between air-delivered and ground-launched systems.

Protocol IV

Blinding Laser Weapons (13 October 1995):
Protocol IV prohibits the use and transfer of laser weapons specifically designed, as their sole combat function or as one of their combat functions, to cause permanent blindness to unenhanced vision. Protocol IV is widely regarded as a successful example of pre-emptive regulation of an emerging technology.

Protocol V

Explosive Remnants of War (28 November 2003):
Protocol V addresses the post-conflict humanitarian impact of explosive remnants of war (unexploded or abandoned explosive devices). It established obligations regarding the clearance, removal or destruction of explosive remnants of war; the recording, retention and transmission of information; the protection of civilian population; and international cooperation and assistance.[80]

Ottawa Treaty

Officially known as the Convention on the Prohibition of the Use, Stockpiling, Production and Transfer of Anti-Personnel Mines and on their Destruction, the Ottawa Treaty is a 1997 convention drafted with the goal of eliminating anti-personnel mines all around the world. The convention defines an anti-personnel mine as "a mine designed to be exploded by the presence, proximity or contact of a person and that will incapacitate, injure or kill one or more persons".[81]

The first article of the treaty prohibits member nations from using, developing, acquiring or even assisting in the production of anti-personnel mines. Through articles IV and V, the convention also mandates its members to destroy existing stockpiles of anti-personnel mines, including those in already mined areas. The further articles of the convention create mechanisms for collaborative efforts towards achieving the goal

[80] United Nations Office for Disarmament Affairs. (n.d.). *The convention on certain conventional weapons* . The Convention on Certain Conventional Weapons | United Nations Office for Disarmament Affairs. https://disarmament.unoda.org/en/our-work/conventional-arms/convention-certain-conventional-weapons
[81] *CONVENTION ON THE PROHIBITION OF THE USE, STOCKPILING, PRODUCTION AND TRANSFER OF ANTI-PERSONNEL MINES AND ON THEIR DESTRUCTION* . United Nations. (n.d.-a). https://treaties.un.org/doc/Treaties/1997/09/19970918%2007-53%20AM/Ch_XXVI_05p.pdf

of anti-personnel mine eradication by requiring member nations to provide assistance to fellow member nations in stockpile destruction, the rehabilitation of mine victims and the exchange of data on mine clearance. The convention also contains provisions for multilateral Review Conferences and compliance frameworks.[82]

The Ottawa Treaty was drafted, historically, as a move to ban anti-personnel mines altogether, as opposed to having them regulated under the already existing CCW. The humanitarian threat of mines had become a growing issue, with the ICRC estimating the maiming or killing of 20,000 individuals due to mines by the year 1996. The existing CCW's consensus-based decision-making mechanism was seen as a major limitation by some countries, allowing major military powers to block talks of comprehensive mine bans. The Ottawa Treaty aimed to bypass this limitation by pursuing a blanket ban on anti-personnel mines.[83]

As of October 2026, the Ottawa Treaty has been signed by 131 countries as signatories, enjoying popular support in the United Nations. However, many major military powers such as the United States, China, Russia, India and Pakistan have never joined the convention.[84] Furthermore, Estonia, Poland, Finland, Lithuania and Latvia exited the convention in early 2026, with Ukraine depositing a letter of withdrawal but being prevented from doing so due to being under active armed conflict.

This exemplifies the existing geopolitical deadlock between countries over the question of completely banning anti-personnel mines as opposed to simply regulating them. It also displays how nations' stances on the issue have changed with varying levels of threat to regional stability and national security.[85]

Convention on Cluster Munitions (CCM)

Adopted in May 2008, the Convention on Cluster Munitions (CCM) is a multilateral convention prohibiting the use and development of cluster munitions. It defines the term 'cluster munition' as "a conventional munition that is designed to disperse or release explosive submunitions each weighing less than 20 kilograms, and includes those explosive submunitions".

The convention requires its members to collaborate on stockpile destruction and clearance of cluster munition remnants. It also contains articles mandating members to collaborate on victim rehabilitation and assistance.[86]

However, the CCM faces prominent implementation issues similar to the Ottawa Treaty, with the non-participation of the world's major military powers such as the United States, China, Russia, India and Pakistan, limiting the actual reach of the CCM's efforts. Moreover, the CCM also endured its first-ever withdrawal in March 2025, when Lithuania opted out of the treaty, citing the need for deterrence measures

[82] *CONVENTION ON THE PROHIBITION OF THE USE, STOCKPILING, PRODUCTION AND TRANSFER OF ANTI-PERSONNEL MINES AND ON THEIR DESTRUCTION* . United Nations. (n.d.-a). https://treaties.un.org/doc/Treaties/1997/09/19970918%2007-53%20AM/Ch_XXVI_05p.pdf

[83] *Why the Mine Ban Convention was worth fighting for and still is*. International Committee of the Red Cross. (2020, November 30). https://www.icrc.org/en/document/why-mine-ban-convention-was-worth-fighting-and-still

[84] United Nations. (n.d.). Convention on the Prohibition of the Use, Stockpiling, Production and Transfer of Anti-Personnel Mines and on their Destruction. United Nations. https://treaties.un.org/pages/ViewDetails.aspx?chapter=26&clang=_en&mtdsg_no=XXVI-5&src=TREATY#11

[85] *Five European states withdraw from Mine Ban Treaty*. Human Rights Watch. (2025, July 1). https://www.hrw.org/news/2025/07/01/five-european-states-withdraw-from-mine-ban-treaty

[86] *CCM/77 30 May 2008 diplomatic conference for the adoption of a convention*. Convention on Cluster Munitions. (2008). https://www.clusterconvention.org/files/convention_text/Convention-ENG.pdf

following Russia's invasion of Ukraine.[87] Currently at 123 members committed to the convention, the CCM stands as an alternative to the CCW as a legal instrument seeking a blanket ban on cluster munitions, rather than mere regulation.[88]



A bomb disposal expert works to dismantle a Chilean-made CB-250K cluster bomb in 2009.

Source: Arms Control Association

CHALLENGES

Geopolitical Deadlock

The consensus-based decision-making process that characterises the CCW, while ensuring the participation of major military powers, has sometimes constrained the ability of High Contracting Parties to adopt substantial measures in response to emerging humanitarian concerns. Different national security priorities, differing assessments of military necessity, and broader geopolitical tensions have frequently interrupted progress on the negotiations of new Protocols or strengthening already existing Protocols.

This dynamic has contributed to a perception among some States and civil society actors that the Convention risks losing relevance in the face of rapid technological change and evolving patterns of armed conflict. However, the requirement of consensus remains highly valued by many High Contracting Parties as a safeguard against any unfair obligation that does not match existing military and defence strategies.

Balancing the imperative of effective humanitarian regulation with the need to maintain broad participation, political legitimacy, and the confidence of all High Contracting Parties continues to constitute one of the central and enduring challenges facing the CCW framework.

LAWS Integration

The integration of emerging technologies in the area of lethal autonomous weapons systems (LAWS) into the existing legal architecture of the CCW presents a complex and ongoing challenge. While the Group of Governmental Experts has made significant progress in clarifying the application of international humanitarian law to such a system, and in identifying key elements related to human judgment, control and substantial differences among High Contracting Parties. Doubts continue to form concerns based on the precise characterisation of lethal autonomous weapon systems (LAWS), the allocation of responsibility and accountability and whether a combination of prohibitions and positive obligations would be appropriate.

The rapid pace of technological development, including advances in artificial intelligence and machine learning, further complicates efforts to enhance legal norms that will remain relevant as capabilities evolve.

[87] Wareham, M. (2024, August 6). *Lithuania leaving cluster munition ban undermines agreement, threatens crucial norms*. Human Rights Watch. https://www.hrw.org/news/2024/08/01/lithuania-leaving-cluster-munition-ban-undermines-agreement-threatens-crucial-norms

[88] *States Parties and Signatories by region.* Convention on Cluster Munitions. (n.d.). https://www.clusterconvention.org/states-parties/

Technological and Financial Gaps

Significant disparities exist among the High Contracting Parties with respect to technological capacity, financial resources and institutional expertise. These disparities affect the ability of States to implement existing obligations under the CCW and its Protocols, and to contribute to the progressive development of the legal framework.

Many States, particularly those with limited financial resources, face challenges in fulfilling reports such as national legal reviews of new weapons and means of warfare and providing adequate and sustainable victim assistance. The increasing technical complexity of issues, especially those related to autonomy, artificial intelligence, and multi-domain operations, risks widening these gaps further.

Without sustained and adequately resourced efforts to enhance technical assistance and the transfer of knowledge, there is a danger that discussions within the CCW may become dominated by a relatively small number of technologically advanced States.

Such an outcome would undermine the inclusive character of the Convention, weaken the sense of collective ownership of its outcomes, and potentially reduce the legitimacy and effectiveness of any new measures that may be adopted.

Regulatory Inadequacy

Certain features of the existing Protocols have given rise to questions regarding their continued adequacy in light of contemporary operational realities and technological developments. The design-based definition of incendiary weapons contained in Protocol III, for example, has been the subject of sustained discussion, with a number of High Contracting Parties and observers noting that multipurpose munitions capable of producing severe burn effects may fall outside its formal scope.

Similarly, the provisions of Amended Protocol II and Protocol V, while constituting important advances at the time of their adoption, face significant implementation challenges in complex and urbanised conflicts characterised by the widespread use of improvised explosive devices and the generation of large quantities of explosive remnants of war. More broadly, the modular structure of the Convention, which permits States to accept individual Protocols selectively, has resulted in uneven levels of legal obligation across the international community.[89]

While this flexibility has encouraged wider participation and allowed the Convention to attract States with differing security perspectives, it has also limited the emergence of a universally applicable regulatory regime.

IMPLICATIONS

Multi-Domain Warfare

The evolution of multi-domain operations, which seek to integrate capabilities across land, maritime, air, space and cyber, carries significant implications for the continued relevance and effective implementation of the CCW. Contemporary military doctrines emphasise the coordinated use of forces and systems across multiple domains, thereby blurring the traditional distinction between weapons and their categories.[90]

[89] United Nations Office for Disarmament Affairs. (n.d.-aa). Framework of the CCW. ional-arms/convention-certain-conventional-weaponshttps://disarmament.unoda.org/en/our-work/convent

[90] International Review of the red cross. (n.d.-c). Evolution of multi-domain operations and blurring of traditional distinctions. https://international-review.icrc.org/articles/reports-and-documents-ihl-and-the-challenges-of-contemporary-armed-conflicts-927

In this evolving environment, munitions and systems regulated under the existing Protocols of the CCW may be employed with emerging technologies such as autonomous systems, DEWs, and operational ideas that were not conceptually anticipated at the time of the CCW's adoption or subsequent amendment. The shortened chain of decision-making and cross-domain communication of multi-domain operations may also complicate efforts to ensure meaningful human judgement and control, particularly with the emerging technology of autonomous systems.[91]

Furthermore, the recording, retention and transmission of information required under Protocol V as well as victim assistance obligations may become more challenging when weapons effects extend across multiple domains.

Frequent Violations of the CCW

Reports and allegations concerning the use of weapons restricted or regulated under the CCW in contemporary armed conflicts have highlighted persistent challenges relating to compliance, transparency and enforcement. Concerns have been raised in various international forums regarding the employment of incendiary weapons in or near concentrations of civilians and the use of anti-personnel mines and IEDs in ways that contravene the provisions of the Amended Protocol II.

The generation of significant quantities of explosive remnants of war without the timely implementation of clearance and victim-assistance measures required under Protocol V. These developments draw attention to the limitation inherent in the Convention's existing architecture, which relies heavily on national implementation measures, the submission of annual reports and cooperative procedures rather than performing independent verification or enforcement mechanisms.[92][93]

While the Convention can provide consultation and cooperation, the absence of formal fact-finding or accountability procedures can impede effective responses to alleged violations. In addition, certain definitional features of the Protocols, notably the definition of incendiary weapons contained in Protocol III, have contributed to interpretive difficulties, particularly with respect to multipurpose munitions that may produce severe burn effects while remaining outside the formal scope of the Protocol.[94]

Formation of Localised Conventions

The emergence of specialised instruments outside the framework of the CCW, notably the 1997 Ottawa Treaty and the 2008 CCM, reflects a pattern in which coalitions of States have pursued more detailed prohibitions when a majority within the CCW has proven difficult to achieve.

These instruments have delivered significant humanitarian benefits and achieved broad participation. At the same time, they have produced a more fragmented legal landscape in which the scope of legal obligations differs between States Parties to the specialised conventions and those that remain outside them or have chosen to adhere to particular Protocols under the CCW.

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[91] *Report of the 2019 session of the Group of Governmental Experts on Emerging Technologies in the Area of Lethal Autonomous Weapons Systems*. (2019). https://documents.unoda.org/wp-content/uploads/2020/09/CCW_GGE.1_2019_3_E.pdf

[92] *Final document: Second Review Conference of the States Parties to the Convention on Prohibitions or Restrictions on the Use of Certain Conventional Weapons Which May Be Deemed to Be Excessively Injurious or to Have Indiscriminate Effects* (CCW/CONF.II/2) [PDF]. United Nations.

[93] Canada. (2025). Working paper (CCW/MSP/2025/WP.12) [Working paper]. United Nations.

[94] International Humanitarian Law Databases. (n.d.-n). Definition of Incendiary Weapons. https://ihl-databases.icrc.org/en/ihl-treaties/ccw-protocol-iii-1980/article-1

Recent decisions by a number of States to withdraw or to reconsider their commitments under these specialised conventions often cite changes in the regional security environment which illustrate the tensions between humanitarian disarmament and evolving national defence assessments. These developments carry implications for the coherence and resilience of global regimes governing conventional weapons. They risk creating inequality in the application of norms among States that cooperate closely in the military plane.

Ecological Degradation

One of the major threats to human security comes from ecological degradation. A few significant factors include deforestation, groundwater depletion, and pollution.

The primary reason for military activities being ignored as a major contributor to environmental degradation is that it's not seen as an industry. Military forces alone account for about two-thirds of CFC-113 ozone depletion. Furthermore, the reversal of environmental pollution damage is approximately USD 500 billion.[95] The physical environment is caught in the crossfire of militarised conflicts because, more often than not, defence ministries are mandated to resolve issues at the cost of destabilising the environment.[96]

Since 1999, the United Nations Environment Programme(UNEP) has conducted about 20 evaluations, assessing the implications of war for the environment. Moreover, on 8 July 2019, 28 draft legal principles were adopted by the International Law Commission(ILC) to acknowledge and manage the consequences of militarised conflict on the environment.[97]

CONCLUSION

The Convention on Certain Conventional Weapons and its annexed Protocols remain a cornerstone of international efforts to prohibit and restrict the use of weapons that may cause superfluous injury or have indiscriminate effects. Over more than four decades, the Convention has demonstrated its capacity to adapt to emerging humanitarian concerns through adaptive progression.

At the same time, rapid advances in military technology, particularly in the field of autonomy and artificial intelligence, pose new questions regarding the continued adequacy of existing Protocols. As the international community prepares for the Seventh Review Conference, Member States are encouraged to assess the implementation of the Convention and its Protocols, to consider ways of enhancing universalisation and compliance, and to examine whether further measures may be required to ensure that technological progress remains consistent with the principles of distinction, proportionality and the protection of the civilian population.

[95]The military's impact on the environment: (n.d.-a). https://www.ipb.org/wp-content/uploads/2017/03/briefing-paper.pdf

[96]Full article: Socio-ecological disruptions in a militarized landscape: The unfolding nature of the Rohingya crisis in Bangladesh. (n.d.-b). https://www.tandfonline.com/doi/full/10.1080/13562576.2025.2576892

[97]*Curbing negative environmental impacts of war and armed conflict*. UNEP. (n.d.). https://www.unep.org/en/news-and-stories/statements/curbing-negative-environmental-impacts-war-and-armed-conflict